

Assignment – 7.2

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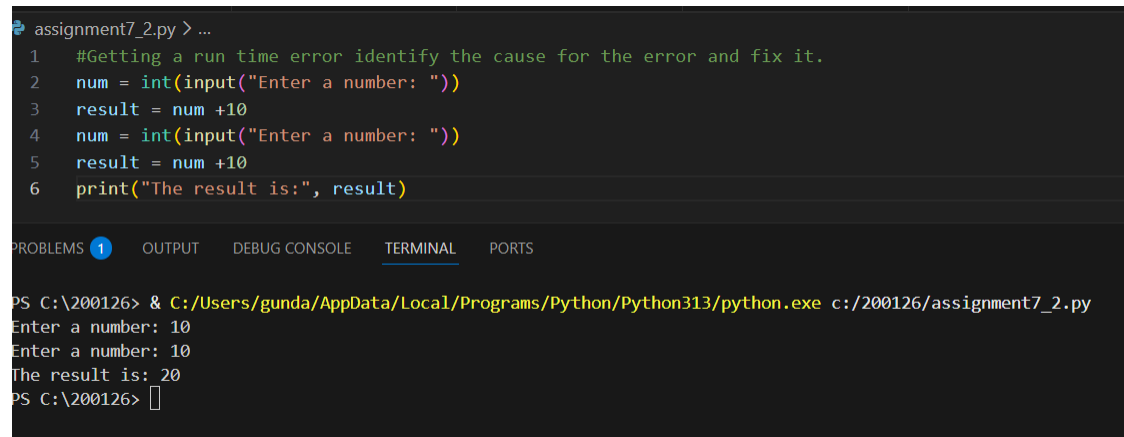
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Task 1 – Runtime Error Due to Invalid Input Type

PROMPT : Getting a run time error identify the cause for the error and fix it.

CODE and OUTPUT:



```
assignment7_2.py > ...
1  #Getting a run time error identify the cause for the error and fix it.
2  num = int(input("Enter a number: "))
3  result = num +10
4  num = int(input("Enter a number: "))
5  result = num +10
6  print("The result is:", result)

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\200126> & C:/Users/gunda/AppData/Local/Programs/Python/Python313/python.exe c:/200126/assignment7_2.py
Enter a number: 10
Enter a number: 10
The result is: 20
PS C:\200126> 
```

Justification : AI helped me understand that the input() function in Python returns a string by default, which caused a TypeError when performing arithmetic operations. By converting the input into an integer using type casting, the runtime error was resolved successfully. This task improved my understanding of data types and their importance in arithmetic operations. It also increased my confidence in identifying and fixing type-related runtime errors

Task 2 Incorrect Function Return Value

PROMPT : Not getting an output identify the cause for the error and fix it.

CODE and OUTPUT:

```
8  #Not getting an output identify the cause for the error and fix it.
9  def square(n):
10     result = n * n
11  def square(n):
12     result = n * n
13     return result
14  number = 5
15  squared_value = square(number)
16  print(f"The square of {number} is {squared_value}.")
17  |
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\200126> & C:/Users/gunda/AppData/Local/Programs/Python/Python313/python.exe c:/200126/assignment7_2.py
Enter a number: 10
Enter a number: 10
The result is: 20
PS C:\200126> ^C
PS C:\200126> & C:/Users/gunda/AppData/Local/Programs/Python/Python313/python.exe c:/200126/assignment7_2.py
The square of 5 is 25.
PS C:\200126> |
```

Justification : AI assisted in identifying that the function was missing a return statement, which caused it to return None instead of the expected value. By adding the appropriate return statement, the function was able to correctly provide the computed result. This helped me better understand how functions work internally and why returning values is important. It strengthened my knowledge of function structure and behavior in Python.

Task 3 IndexError in List Traversal

PROMPT : Getting an index out of range error identify the cause for the error and fix it

CODE and OUTPUT:

```
18 # Getting an index out of range error identify the cause for the error and fix it
19 numbers= [1, 2, 3]
20 for i in range(len(numbers)):
21     print(numbers[i])
22 numbers= [1, 2, 3]
23 for i in range(len(numbers)):
24     print(numbers[i])
25
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\200126> & C:/Users/gunda/AppData/Local/Programs/Python/Python313/python.exe c:/200126/assignment7_2.py
Enter a number: 10
Enter a number: 10
The result is: 20
PS C:\200126> ^C
PS C:\200126> & C:/Users/gunda/AppData/Local/Programs/Python/Python313/python.exe c:/200126/assignment7_2.py
The square of 5 is 25.
PS C:\200126> & C:/Users/gunda/AppData/Local/Programs/Python/Python313/python.exe c:/200126/assignment7_2.py
1
2
3
1
2
3
PS C:\200126> 
```

Justification : AI detected that the loop boundary condition exceeded the valid index range of the list, which caused an IndexError. By correcting the loop range and removing the extra increment, the program executed without errors. This task helped me understand the importance of proper indexing and boundary conditions when working with lists. It also improved my debugging skills related to iteration logic

Task 4 Uninitialized Variable Usage

PROMPT : Getting a name error identify the cause for the error and fix it.

CODE:

```
45 if True:
46     pass
47 print(total)
48 #Getting a name error identify the cause for the error and fix it.
49
50 #The error occurs because the variable 'total' is not defined anywhere in the code. To fix this, you need to define the variable 'total' before trying to print it.
51 #Here's an example of how to fix the code:
52 total = 0 # Define the variable total before using it
53 if True:
54     pass
55 print(total) # Now this will print 0 without any errors
56
```

OUTPUT:

0

PS D:\AI>